UNIT 2 · TOPIC 1(D)

Gears

Syllabus topic: Design and selection of machinery elements: gears, pulleys, chains and sprockets, belts

Law of gearing

For a constant velocity ratio, the common normal at the point of contact must pass through the pitch point. The involute profile satisfies this and tolerates small changes in centre distance.

Terminology

Term	Definition
Module	m = D/T (mm per tooth) — Indian/ISO standard sizing
Diametral pitch	Teeth per inch — US practice
Circular pitch	$p_c = \pi m$
Addendum / dedendum	m / 1.25 m (20° full-depth)
Backlash	Play between mating tooth flanks — needed for lubrication and thermal expansion

- **Pressure angle:** 20° full-depth is standard (14.5° is obsolete). A higher pressure angle gives a stronger tooth and fewer minimum teeth, but more bearing load.

$$\text{Minimum teeth to avoid interference} = 2/\sin^2\varphi \approx 17\text{–}18 \text{ for } \varphi = 20° \ (32 \text{ for } 14.5°)$$

Gear types

Type	Features / use
Spur	Parallel shafts; no axial thrust
Helical	Quieter; axial thrust
Herringbone / double-helical	Thrust cancelled
Straight & spiral bevel	Intersecting shafts — the tractor differential crown-pinion is spiral bevel or hypoid
Hypoid	Offset axes — allows a lower drive line
Worm & wheel	Large reduction, 10–100:1 in one stage; can be self-locking; low efficiency
Rack & pinion	Steering

Tooth strength — Lewis equation

$$F_t = \sigma_b \cdot b \cdot y \cdot p_c = \sigma_b \cdot b \cdot Y \cdot m$$

- The equation treats the tooth as a cantilever. Design the weaker of pinion and gear (lower σ·Y).
- Follow with dynamic (Buckingham) and wear-load checks.

Gear trains

- **Simple train:** intermediate (idler) gears change direction only.
- **Compound train:** VR = product of the stage ratios.
- **Epicyclic/planetary trains** give large ratios compactly (tractor final drives, planetary steering).

Materials

C45/alloy steels (hardened); cast iron (low speed); nylon/plastics (light drives, seed-metering gears).

Practice MCQs

Q21. The standard pressure angle for involute gears today is:

(a) 14.5° (b) 20°
(c) 25° (d) 30°

Q22. Minimum number of teeth on a 20° full-depth pinion to avoid interference is about:

(a) 12 (b) 17
(c) 25 (d) 32

Q23. Module of a gear equals:

(a) T/D (b) D/T
(c) πD/T (d) T/πD

Q24. The gear pair connecting shafts whose axes are offset (non-intersecting, non-parallel), used in automotive/tractor drive lines, is:

(a) Spiral bevel (b) Hypoid
(c) Herringbone (d) Rack and pinion

Q25. A worm-and-wheel drive is preferred where:

(a) High efficiency is needed (b) Large speed reduction in one stage (and possible self-locking) is needed
(c) Shafts are parallel (d) Axial thrust must be zero

Q26. In the Lewis equation $F_t = \sigma \cdot b \cdot Y \cdot m$, Y is the:

(a) Wear factor (b) Lewis form factor (tooth shape)
(c) Service factor (d) Velocity factor

Q27. An idler gear in a simple gear train:

(a) Changes speed ratio (b) Changes only the direction of rotation
(c) Increases torque (d) Reduces noise

Answers

Q21 — (b) 20°
Q22 — (b) 17 — *Minimum teeth = $2/\sin^2 20° \approx 17.1 \rightarrow 17–18$.*
Q23 — (b) D/T
Q24 — (b) Hypoid
Q25 — (b) Large speed reduction in one stage (and possible self-locking) is needed
Q26 — (b) Lewis form factor (tooth shape)
Q27 — (b) Changes only the direction of rotation

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 4 (Elements of Rotary Power-Transmission Systems) and Appendix (ASAE V-belt standards); standard machine design references (Shigley; V.B. Bhandari); ASAE/IS standards.